

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

The Importance of Measur



Learning Outcome:

Make the coding of the system modules and the main program, starting from using the chosen programming language, based on design specifications.
Exchanges information about requirements types and their characteristics.

Can Do Statements:

- I can talk about requirements measures.
- I can describe quality requirements.
- I can explain what domain requirements are.

Ask your partner:

- Do you tend to think about your personal performance as a system professional? Why?
- How important is it for software organizations to analyze past projects? Why?

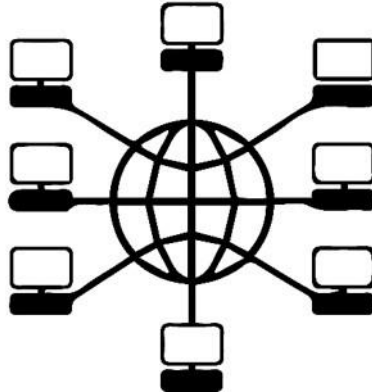




Let's Explore!

Skipping class*...

*The expression "skip class" is used when you do not attend class.



Instructions: Read the text below.

- Jessica:** Hi, Steven. I have got to find someone who can tell me about the topic the teacher covered yesterday.
- Steven:** The teacher talked about the requirements measures. You might want to understand this subject thoroughly before next class.
- Jessica:** Requirements, what?
- Steven:** The requirements measures. All software organizations had better keep track of the statistics of the products they produce as they provide important information that can improve and strengthen the production phases of future projects. You might want to check out different metric systems such as product size, which assesses time, money, speed and portability; the requirements status, which indicates the level of achievement of the requirements; the request for changes; and effort.
- Jessica:** Now that I think about that, the quality requirements are also important within those metric systems. Their correct implementation is going to have to provide objective data about the faulty requirements, highlighting those that were defective, unnecessary and unfinished. Software organizations ought to analyze these pieces of information as they unveil the difficulties that the different areas of the company are presenting. Furthermore, we ought to obtain, from these reports; valuable information related to the costs that flaws represent to the company.
- Steven:** That is correct. The last thing you have got to know about yesterday's class is the domain requirements. They basically display, in specialized terminology, the application domain of the system; in other words, the environment in which the system runs. Finally, they had better indicate the basics of the application domain in order to make the system work in correspondence to its goals.
- Jessica:** Thank you for the class update. I am going to have to invite you for a coffee after all this help.

**Let's Practice!**

Jessica is talking about requirements measures advantages for a project.

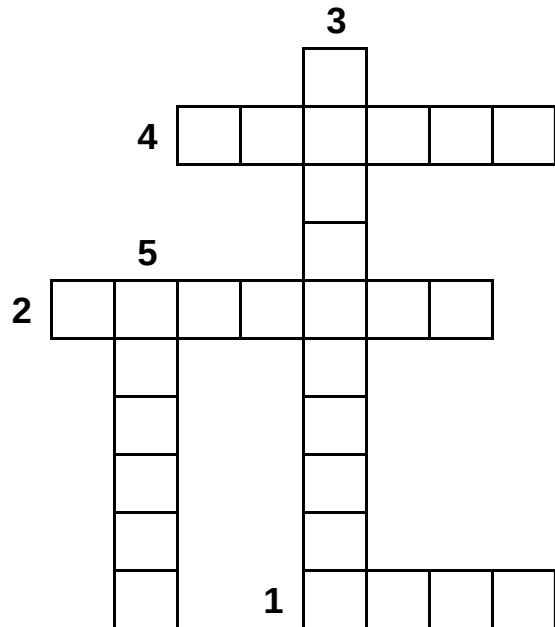
Instructions: Read the clues. Complete the crossword.

Down

3. The manager is going to have to go over the data and _____ gathered to plan improvements for their next project.
5. Requirements status have got to _____ the progress for all tasks.

Across

1. The product _____ has got to be determined by analyzing elements such as overall effort, time and money.
2. _____ requirements contain all the criteria about defects that system is going to have to process and classify.
4. You have got to check out different _____ systems such as product size, which assesses time, money, speed and portability.

**Let's Practice!**

Steven recognizes the importance of quality requirements measures.

Instructions: Circle the correct answer to complete the paragraph.

You (might want / might want to) go over quality requirements before next session since they are very important for metric systems. You (ought to / ought for) implement them correctly in order to provide objective data about the faulty requirements, highlighting those that were defective, unnecessary and unfinished. Software organizations (have better / had better) analyze these pieces of information as they unveil the difficulties that the different areas of the company are presenting. Furthermore, thanks to these reports, we (ought to obtain / ought to obtains) valuable information related to the costs that flaws represent to the company. So, you (may wanted / might want to) borrow my notes if you want to study quality requirement.





Extension Activity

The importance of know the scope and measures for a project

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Imagine that you are the director of a software company.
- Look for information about quality requirements measures.
- Design a brochure to show the importance and how to apply those measures in a project.
- Make a presentation to your classmates and answer all the questions.

Self-Assessment

Yes

No

- I look for information about quality requirements.
- I write about importance and application of measures in software projects.
- I talk about quality measures for projects.

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Target Words

- **statistic** (noun): information that comes out of the analysis of real situations, projects or events
e.g. All software organizations had better keep track of the statistics of the products they produce as they provide important information that can improve and strengthen the production phases of future projects.
- **metric system** (noun): a method that is useful to measure and assess something
e.g. You might want to check out different metric systems such as product size, which assesses time, money, speed and portability.
- **portability** (noun): the attribute of an object that is portable or applicable for different objectives
e.g. You might want to check out different metric systems such as product size, which assesses time, money, speed and portability.
- **objective** (adjective): real, not influenced by the personal perception
e.g. Their correct implementation is going to have to provide objective data about the faulty requirements, highlighting those that were defective, unnecessary, and unfinished.
- **faulty** (adjective): which is not perfect, that has plenty of mistakes
e.g. Their correct implementation is going to have to provide objective data about the faulty requirements, highlighting those that were defective, unnecessary, and unfinished.
- **unveil** (verb): to expose or show something to others
e.g. Software organizations ought to analyze these pieces of information as they unveil the difficulties that the different areas of the company are presenting.



Learning Points

Have Got to for What Is Necessary

Have got to represents the idea that something is necessary or a must.

When you use **have got to**, you are saying that something has to be done and there is no other possible action to take in the present time.

This is the structure you are going to follow:

Subject + **have/has** + **got to** + **verb (base form)** + complement.

For example,

*Functional requirements **have got to** show the behavior of the system.*

*Business requirements **have got to** illustrate the objectives or visions of the system.*

Examples

- *I have got to find someone who can tell me about the topic the teacher covered yesterday.*
- *The last thing you have got to know about yesterday's class is the domain requirements.*



Learning Points

Requirements Measures

Each system or software project that is developed has to be measured. Its metrics help software organizations to keep important information of what they produce.

There are some elements that are considered to be the most important metric systems that a software organization can carry on with their projects.

PRODUCT SIZE

The most essential metric has to deal with the amount of requirements which contain valuable information on the way the system should behave in certain situations. The size of the product is obtained by analyzing such elements like the overall effort, time and money that it took to develop the complete system. Furthermore, aspects of the non-functional requirements, such as safety and the general parameters that the system has to accomplish have to be included within the product size estimation.

REQUIREMENTS QUALITY

Having an objective report of the quality of the requirements provides useful data to the software organizations. Information about the defective, unnecessary, unfinished and missing requirements on past projects reflect the inner flaws of the organization by reporting the logistic and technical issues that have to be addressed during the stages of design, coding and testing.

Once the information that was collected is analyzed, it can be grouped into two different categories: Efficiency and effectiveness. The first one, making reference to the average number of flaws detected during the inspection process reflecting the cost that finding defects has. On the other hand, effectiveness makes reference to the amount of flaws detected in the final product during the inspection stage, reflecting how good this stage is being performed.

REQUIREMENTS STATUS

Keeping track of the status of the projects' requirements is vital to determine the progress it has. Different status give information about the level of completion of the project: Rejected, deleted, deferred (to be implemented in the future), verified (passed all the different tests after its integration), implemented, approved, and proposed (it was recommended by someone).

REQUEST FOR CHANGES

The record of any type of modification or change in the project's requirements provides information on the way these are addressed. The amount of changes, the moment that those were found and proposed and the time it took to implement them, provide an insight of all the organs that intervene in the creation of the project and can show what processes in which branches of the organization need to be revised.

EFFORT

Knowing how much effort it took employees to successfully complete a project can give important evidence for future projects.

